Unit: Unit 0: Foundations of Algebra
Topic: Natural Numbers Integers

Name: ______________
Date: __________

Foundation (Jalapeno)

Q1. Classify the number 5 as:
(A) Natural and whole
(B) Natural but not whole
(C) Whole but not natural
(D) Integer but not natural
(E) None of the above

Q2. Identify the next even number in the pattern: $2, 4, 6, 8, 10$
(A) 12
(B) 14
(C) 16
(D) 18
(E) 20

Q3. A basketball player scored -3 points in a game due to penalties. If they scored 5 points from shots, what is their total score?
(A) 2
(B) 5
(C) -3
(D) 8
(E) -8

Q4. A gymnast performs 15 routines. If 9 routines are odd-numbered, how many are even-numbered?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q5. The temperature at a sports stadium is -2 degrees Celsius. If it rises by 5 degrees, what is the new temperature?
(A) 3
(B) 5
(C) 7
(D) -2
(E) -7

Q6. A cyclist travels 12 km in 2 hours. How many kilometers do they travel per hour?
(A) 4 km/h
(B) 5 km/h
(C) 6 km/h
(D) 8 km/h
(E) 10 km/h

Q7. A sports team wins 8 games and loses -2 games. What is their total number of games played?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 14

Q8. Identify the pattern in the sequence: $1, 3, 5, 7, 9$
(A) Adding 1
(B) Adding 2
(C) Adding 3
(D) Multiplying by 2
(E) Adding 2 then subtracting 1

Open Ended Questions (FRQ)

Q9. A marathon runner travels 25 km in 5 hours. If they travel at a constant speed, how many kilometers will they travel in 8 hours? Show your calculations and explain your reasoning.

Q10. A swimmer finishes a race in 2 minutes and 15 seconds. If their opponent finished the same race in 2 minutes and 30 seconds, how many seconds faster did the swimmer finish? Convert all times to seconds and show your work.

Chef's Tips

- Emphasize the importance of showing work and calculations for free-response questions
- Encourage students to use real-world examples to illustrate their understanding of natural numbers and integers
- Remind students to check their units and ensure they are answering the question being asked